

Assignment – Module 1: Information Retrieval

Total Marks: 30

Instructions

- Write your Name and Roll No. on the first page.
- Take clear photographs/scans of all pages and combine them into one PDF.
- Upload the PDF on Google Classroom.
- Also submit the handwritten assignment.
- Use the prescribed book as the primary reference. PPTs are only for understanding.
- Follow the book's notations, terminology, and standard conventions.
- Show all intermediate steps wherever required.

Q1. Consider the following document collection:

6 Marks

- D1: information retrieval uses indexing
- D2: indexing improves information search
- D3: information search uses efficient retrieval

(a) Construct the inverted index for the collection. [3]

(b) Using your index, process the Boolean query:

information AND retrieval [2]

Show the postings-list operations involved.

(c) Explain how the query could be processed more efficiently. [1]

Q2. Consider:

5 Marks

- D1: information retrieval systems process information
- D2: information systems support retrieval
- D3: efficient information retrieval

(a) Construct the positional index for **information, retrieval, and systems**. [3]

(b) Demonstrate how the phrase query:

"information retrieval"

is processed using the positional index. [2]

Q3. Consider the wildcard query:

6 Marks

re*ve

- (a) Show how this query is processed using a Permuterm index. [2]
 - (b) Using a 3-gram index, add the boundary symbol \$ and derive the Boolean k-gram query. [2]
 - (c) Explain why candidate terms obtained from a k-gram index require post-filtering before retrieving their postings. [2]
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Q4. Suppose the user enters the misspelled term:

5 Marks

retrival

and the vocabulary contains:

retrieval, removal, arrival, retrievals

- (a) Explain how k-gram overlap can be used to generate candidate corrections. [2]
 - (b) Calculate the edit distance (Levenshtein) between **retrival** and **retrieval**. [2]
 - (c) Explain why candidate generation should be performed before calculating edit distance for every vocabulary term. [1]
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Q5. A search engine needs to index a collection that is too large to fit completely in main memory.

8 Marks

- (a) Describe how **BSBI (Blocked Sort-Based Indexing)** can be used to construct the inverted index. Show the major steps. [3]
- (b) Explain how **SPIMI (Single-Pass In-Memory Indexing)** would construct the same index and identify one major difference from BSBI. [3]
- (c) If the collection is continuously updated, explain how dynamic indexing can be used to handle new documents efficiently. [2]